Finite Pulse Certificates and Shrinking-Width Schedules for a Local Immersion Obstruction

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Abstract

We study the Darboux-height mechanism in the assigned smooth-metric construction. A finite oscillatory testing inequality retains the actual forcing residual, terminal-data norm, coefficient derivatives and physical hyperbolic energy. It leads to a one-parameter pulse schedule with width $w_\tau = \tau^{-a}$, $1 < a < 2$, in place of first fixing a small constant times τ^{-1}. The two competing relative errors are τw_τ and $(\tau^2 w_\tau)^{-1}$; $a = 3/2$ balances this explicit budget at order $\tau^{-1/2}$. All finite Taylor derivatives and metric-seminorm costs are retained. A quantitative rank-one orientation estimate describes the separate geometric normalization of a hypothetical immersion. The source cap estimate, category argument and smooth two-level assembly are identified as distinct inputs, not consequences of finite jets or numerical checks. Contribution assessment: C3/5, an explicit residual-sensitive pulse obstruction and a rigorously controlled shrinking-width regime give notable local refinements without independently certifying the global nonimmersion theorem.

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1 The exact problem and source dependencies

The source [1] asserts existence of a smooth positive metric g on $(-1,1)^2$, with the complete Euclidean Taylor jet at the origin, such that no neighborhood of the origin admits a C^∞ isometric immersion into $\mathbb{R}^3$. This is not an assertion about a prescribed immersion or a fixed finite differentiability class. Its metric uses both signs of Gaussian curvature; it does not settle the corresponding unrestricted smooth problem with $K \geq 0$ or $K \leq 0$. The present report proves the local results stated below, not an independent certification of that entire assertion.

On $S = [-3,3]^2$, the source fixes

$$K_0(x,y) = \kappa(x^2 - h(y)), \qquad \kappa > 0,$$

where h is smooth and even, equals one on $[-1/2,1/2]$, is nonincreasing for $y > 0$, is positive for $|y| < 1$, negative for $|y| > 1$, and has $h'(1) < 0$. Metrics are perturbed only in $\mathcal{P} = [-1/10,1/10]^2$, retaining $K_g < -\kappa/2$ on $[-1/5,1/5]^2$. The topology is the smooth Fréchet topology; an open neighborhood restricts finitely many seminorms, not all derivative orders with a common bound.

For a height z, put

$$H_{ij} = z_{ij} - \Gamma^r_{ij} z_r, \quad E = \det g\,(1 - g^{ij} z_i z_j), \quad H_{xx}H_{yy} - H_{xy}^2 = K_g E. \tag{1}$$

An admissible height is smooth on a neighborhood of the entire closed square and obeys $E > 0$, $H_{yy} \neq 0$, and $|H_{xy}/H_{yy}| \leq 1/100$ there. Nonexistence of this restricted whole-square height is not pointwise nonsolvability near each point. In particular the source's curvature zeros are clean changes of sign.

If F is an immersion and e a fixed ambient unit vector, then $z = \langle F, e\rangle$ has $H = \langle n, e\rangle\,\mathrm{II}$. Orthogonal decomposition of e gives $1 - |dz|_g^2 = \langle n, e\rangle^2$, and the Gauss equation gives (1). This necessary condition does not require a reconstruction theorem for a scalar solution. At a point where $e = n$, $dz = 0$ and $H = \mathrm{II}$.

1.1 Architecture, gauges and imported estimates

All eleven assigned text files were freshly read, including the four mathematical modules. The main text states the regularity and sign limitations. The model module gives (1). The cap module derives a mixed-type regularity estimate, the pulse module combines two caps with hyperbolic transport and Baire category, and the assembly module passes to arbitrary immersions. The four figure files describe the geometry; they are not additional analytic hypotheses.

The cap theorem assumes uniform C^8 bounds for g, z, a positive metric lower bound, and positive lower bounds for $|H_{yy}|$ and E. Writing

$$q = H_{xy}/H_{yy}, \quad D = \partial_x - q\partial_y, \quad G = E/H_{yy}^2, \quad a = KG,$$

the solution-dependent chart is $x = t$, $y_t = -q$, $y(0,s) = s$. Hence $D = \partial_t$, $|y - s| \leq 1/50$ on the selected chart, and s_y is bounded above and below. The cap theorem gains all ordinary derivatives on compact subcaps from all ordinary metric derivatives there. It imposes no high-order data on the artificial fourth edge.

Its proof combines a directed multiplier with an elliptic estimate. For $d = b - s$, the weights are

$$W = d^8 e^{-\lambda s + \int_0^t B(r,s)\,\mathrm{d}r}, \quad m = -W, \quad n = \epsilon t W.$$

The curvature drift and the tangential correction control $K \leq c_1 d$; a weighted elliptic estimate controls the complement. The choices are ϵ, then λ, then c_1. The artificial edge is inside the individual smooth solution's domain, so vanishing weights remove its boundary terms before uniform estimates

are extracted. The present pulse extension *uses this cap theorem as a source input.* No finite coefficient test establishes it.

The pulse proof fixes one bounded height class, a rational approximation q_0 to $q(0,0)$, and linear coordinates $\xi = x,\ t = y + q_0 x$. On a fixed negative-curvature slab,

$$d_0 = (q - q_0)^2 + a < 0, \quad H_{\xi\xi} = d_0 H_{yy}, \quad H_{\xi t} = (q - q_0) H_{yy}.$$

Solving the equation for z_{tt} gives

$$z_{tt} = Q_g(\xi, t, z_i, z_{\xi\xi}, z_{\xi t}) = \Gamma^i_{tt} z_i + \frac{H^2_{\xi t} + (K \det g)(1 - |dz|^2_g)}{H_{\xi\xi}}. \tag{2}$$

Its principal partial derivatives along a solution are $Q_{z_{\xi t}} = 2(q - q_0)/d_0$ and $Q_{z_{\xi\xi}} = -1/d_0 > 0$. All claims below stay within a compact jet region where the denominator and this hyperbolicity margin remain separated from zero. Constants depend on the fixed patch and bounded class, not uniformly on $\kappa \downarrow 0$ or on arbitrary immersions.

2 Full differentiated rows and finite comparison

The fresh original private proofs [6, 7] retain every coefficient derivative and distinguish polynomial closure from a smooth remainder. For a local differential operator, their exact rule is

$$\partial^\gamma (A_\alpha \partial^\alpha z) = \sum_{\delta \le \gamma} \binom{\gamma}{\delta} (\partial^\delta A_\alpha) \partial^{\alpha + \gamma - \delta} z. \tag{3}$$

An output jet of order r for a second-order equation generally uses input derivatives through $r + 2$, rather than a square closed system at order r.

For the cap equation solved as $z_{xx} = \mathcal{P}$, direct differentiation of its rational expression gives

$$\mathcal{P}_{z_{xy}} = 2q, \quad \mathcal{P}_{z_{yy}} = -q^2 - a, \quad \mathcal{P}_{z_i} = \Gamma^i_{DD} + K \left(\frac{E_{z_i}}{H_{yy}} + \frac{E \Gamma^i_{yy}}{H^2_{yy}} \right).$$

The contraction Γ^i_{DD} does not differentiate D. For $u = \partial^l_y z$, the next-to-top terms in $\partial^l_y \mathcal{P}$ are $l[2q_y u_x - (q^2 + a)_y u_y]$, in addition to its direct linearization. All these are total coefficient derivatives. Using $D^2 = \partial^2_x - 2q\partial_{xy} + q^2 \partial^2_y - (Dq)\partial_y$, the full top row is

$$\left[D^2 + a\partial^2_y + BD + (Dq - \mathcal{P}_{z_x} q - \mathcal{P}_{z_y} + l a_y) \partial_y \right] u = R_l, \quad B = -\mathcal{P}_{z_x} - 2l q_y. \tag{4}$$

The covariant-Hessian commutator supplies $Dq - \mathcal{P}_{z_x} q - \mathcal{P}_{z_y} = K_y G + O(K)$. After the flow change, $A = G s^2_y K$ and the transverse coefficient is $(l + 1)A_s + O_l(A)$. One copy is geometric, and l copies come from differentiating the second-jet coefficients. Dropping coefficient derivatives loses the drift used by the cap proof. The source already contains this identity; we do not claim it as a new result.

For $l \ge 8$, every remaining solution factor has order at most l, and at most one has order $l - 1$ or l: two such factors would consume at least $2(l - 3) > l$ differentiations. The lower factors use the local two-dimensional Sobolev bound through order $l - 2$, the one high factor its L^2 bound. This explains the source induction without assuming high derivatives of its solution-dependent chart. For an approximate equation $z_{xx} - \mathcal{P} = \rho$, the right side of (4) also contains $\partial^l_y \rho$; a bound on ρ alone does not control it.

Lemma 2.1 (Finite time comparison, not an infinite jet solution)**.** *Fix smooth Cauchy data at $t = -w$ for (2), with the required spatial derivatives bounded, and a fixed smooth background metric. On a compact jet region with $|H_{\xi\xi}| \geq c > 0$, a polynomial*

$$z^0(\xi, t) = \sum_{k=0}^{q+1} u_k(\xi)(t + w)^k, \qquad q \geq 1,$$

can be chosen with the same initial z, z_t such that its residual $\rho^0 = z^0_{tt} - Q_{g_*}(J^2 z^0)$ satisfies

$$|\partial_\xi^j \rho^0(\xi, t)| \leq \frac{(2w)^q}{q!} \sup_{|t|\leq w} |\partial_t^q \partial_\xi^j \rho^0(\xi, t)| \tag{5}$$

for every fixed j for which these derivatives are bounded. For uniformly bounded data and small w, the finite suprema are uniform; no convergence of a formal infinite series is used.

Proof. Choose u_0, u_1 from the data. Having found them through u_{k+1}, set

$$(k + 2)(k + 1)u_{k+2} = [T^k]\, Q_{g_*}(\xi, -w + T, J^2 z^0), \quad 0 \leq k < q.$$

The coefficient on the right only involves those already found: the time derivative has order at most one there. Spatial derivatives of the data, and all metric coefficient derivatives, are retained. Reciprocals are expanded by $b_0 v_k = f_k - \sum_{i=1}^k b_i v_{k-i}$, with $b_0 \neq 0$; this is an exact finite coefficient identity. The residual and its first $q - 1$ time derivatives vanish at the initial edge. Taylor's integral remainder gives (5), also after the declared spatial derivative. Finite coefficient bounds and continuity keep the denominator away from zero throughout a sufficiently short strip. □

3 A finite physical-energy pulse certificate

Use real coordinates (ξ, t), $-w \leq t \leq w$. Spatial intervals I_t shrink at speed v_{edge} at both ends, and contain the support of a fixed nonzero real $\chi \in C_c^\infty(\mathbb{R})$. Let $\phi \in C_c^\infty((-1, 1))$ be nonnegative with $\Phi = \int \phi > 0$. Consider

$$v_{tt} - P v_{t\xi} - S v_{\xi\xi} + b^t v_t + b^\xi v_\xi = F, \quad F = A\tau^2 D\chi(\xi)\phi(t/w)\cos(\tau\xi) + e. \tag{6}$$

Here v is the scalar unknown, distinct from v_{edge}. Assume real C^1 principal coefficients, bounded lower coefficients, $S \geq s_0 > 0$, and a single sign for D on the strip, with $0 < d_0 \leq |D| \leq d_1$. Assume $A > 0$, $\tau > 0$, zero Cauchy data at $t = -w$, and smoothness sufficient for the following integrations. All norms of coefficients are over the moving strip.

Put $\mu = \min(1, s_0)$ and define

$$\begin{aligned} c_E &= \|P_\xi\|_\infty + 2\left\|b^t\right\|_\infty + \|S_t\|_\infty / s_0 + \left\|S_\xi + b^\xi\right\|_\infty / \sqrt{s_0}, \\ H_w &= e^{c_E w}/\sqrt{\mu}, \qquad \epsilon = \int_{-w}^{w} \|e(\cdot, t)\|_{L^2(I_t)}\, \mathrm{d}t. \end{aligned} \tag{7}$$

For $\psi = \chi\cos(\tau\xi)$, let

$$K_\tau = \sup_t(\|(P\psi)_\xi\|_2 + \|(S\psi)_\xi\|_2 + \left\|b^t\psi\right\|_2 + \left\|b^\xi\psi\right\|_2), \qquad J_\tau = \int \chi^2 \cos^2(\tau\xi)\, \mathrm{d}\xi.$$

These are the physical spatial $L^2(\,\mathrm{d}\xi)$ norms, not unweighted coefficient norms of a polynomial representation.

Theorem 3.1 (Finite residual obstruction)**.** *If*

$$v_{\text{edge}} \geq \sup \frac{|P| + \sqrt{P^2 + 4S}}{2}, \tag{8}$$

then any solution of (6) *with the stated zero data must satisfy*

$$\begin{aligned} Aw\tau^2 d_0 \Phi J_\tau \leq & \left| \int \psi v_t(\xi, w)\, \mathrm{d}\xi \right| + \|\psi\|_2 \, \epsilon \\ & + 2wK_\tau H_w\big(Aw\tau^2 d_1 \|\chi\|_2 \, \Phi + \epsilon\big). \end{aligned} \tag{9}$$

Moreover,

$$J_\tau \geq \tfrac{1}{2} \|\chi\|_2^2 - \frac{\|(\chi^2)'\|_1}{4\tau}. \tag{10}$$

If $T_m = \left\| \partial_\xi^m (\chi v_t(\cdot, w)) \right\|_1$ *is certified, the terminal term in* (9) *is at most* $\tau^{-m} T_m$*.*

Proof. For $a = v_t$, $b = \sqrt{S}\, v_\xi$, the energy and flux are

$$e_v = (a^2 + b^2)/2, \qquad j_v = (Pa^2 + 2\sqrt{S}\, ab)/2.$$

The eigenvalues of $\begin{pmatrix} P & \sqrt{S} \\ \sqrt{S} & 0 \end{pmatrix}$ are $(P \pm \sqrt{P^2 + 4S})/2$. Its norm gives exactly (8), making the moving-boundary flux nonpositive. A direct product-rule calculation gives

$$(e_v)_t - (j_v)_\xi = -(P_\xi/2 + b^t) v_t^2 + (S_t/2) v_\xi^2 - (S_\xi + b^\xi) v_\xi v_t + F v_t.$$

Since $|v_\xi v_t| \leq e_v / \sqrt{s_0}$, integration yields

$$\mathcal{E}' \leq c_E \mathcal{E} + \sqrt{2\mathcal{E}}\, \|F\|_2 \,.$$

Use $\sqrt{\mathcal{E} + \eta}$, integrate, and let $\eta \downarrow 0$. Zero initial energy implies

$$\sup_t \|(v_t, v_\xi)\|_2 \leq H_w \int_{-w}^{w} \|F\|_2 \, \mathrm{d}t \leq H_w(Aw\tau^2 d_1 \|\chi\|_2 \, \Phi + \epsilon). \tag{11}$$

Integrate (6) against ψ in space and time. Integrating each other second-order term once in space produces $(P\psi)_\xi v_t$ or $(S\psi)_\xi v_\xi$; no coefficient derivative is discarded. Cauchy–Schwarz and (11) bound these and the first-order terms by the last line of (9). The residual costs $\|\psi\|_2 \, \epsilon$. The sign of D, nonnegativity of ϕ, and $d_0 > 0$ give the left side. No convergence of D is needed. Finally use $\cos^2 u = (1 + \cos 2u)/2$ and one integration by parts to get (10); m integrations prove the terminal estimate. □

The exact speed in (8) sharpens the convenient source bound $|P| + \sqrt{S}$, but is a classical flux computation. It is not the main contribution and it supplies no estimate without the positive lower bound for S.

Remark 3.2 (Measured initial defects). For nonzero initial data with energy $\mathcal{E}_-$, replace the parenthesis in (11) by $\sqrt{2\mathcal{E}_-} + Aw\tau^2 d_1 \|\chi\|_2 \, \Phi + \epsilon$, and replace the terminal moment by $|\int \psi[v_t(w) - v_t(-w)]|$. The same proof works. Thus approximate Cauchy matching must be charged explicitly; it is not automatically a Taylor residual.

4 A shrinking-width regime with explicit errors

The finite inequality has a useful consequence beyond substituting a numerical value into the source's small fixed parameter. We derive fresh estimates in the width itself.

Fix a smooth background g_*, the sheared slab and the cutoffs above. Fix a bounded height class as in the source: $\|z\|_{C^8} \le M$, $|H_{yy}| \ge M^{-1}$, $E \ge M^{-1}$, and the rational slope restriction. Throughout this section assume the source cap estimate and its propagation to both pulse edges, with uniform bounds for every fixed spatial derivative of z, z_t. The same conclusions may instead be supplied as independently verified hypotheses for a particular family. They are not inferred from (9).

Consider

$$\widetilde{g}_\tau = g_* + A_\tau \chi(\xi)\phi(t/w_\tau)\cos(\tau\xi)\ \mathrm{d}t^2, \qquad A_\tau = \tau^{-N}. \tag{12}$$

Let g_τ differ from this metric by γ_τ in C^2, and have all the low-order bounds just specified. Off the pulse, assume the metric has uniform bounds at each fixed derivative order. The source's choice $\|g_\tau - \widetilde{g}_\tau\|_{C^\tau} \le \tau^{-\tau}$ is more than sufficient. A statement only about C^2 closeness would not supply the off-pulse cap hypotheses.

Lemma 4.1 (Width-dependent metric forcing)**.** *Assume $0 < w_\tau \le 1$, $\tau \ge 1$, and $A_\tau(\tau + w_\tau^{-1})^2 + \gamma_\tau \to 0$. In particular g_τ and g_* are sufficiently close in C^2 for large τ. Retain the fixed low-order and denominator margins above. Construct z^0 by Lemma 2.1 with $q = N$. For $v = z - z^0$, the mean linearization of Q_{g_*} gives* (6)*, with uniformly bounded coefficients and positive lower bound for S, whenever both second-jet argument vectors stay uniformly close to their common initial value. Moreover*

$$D = \frac{1 - |dz|^2_{g_*}}{2H_{\xi\xi}(g_*, z)} \tag{13}$$

has one sign and uniform positive absolute lower and upper bounds, and the remainder obeys

$$\begin{aligned}\|e\|_\infty \le C\big[&A_\tau(\tau + w_\tau^{-1}) + A_\tau^2\tau^2(\tau + w_\tau^{-1}) \\ &+ \gamma_\tau + w_\tau^N\big].\end{aligned} \tag{14}$$

Here C depends on finitely many fixed background, cutoff and endpoint bounds. In particular it is independent of τ along families satisfying these uniform hypotheses.

Proof. For a positive metric in the linear coordinates, the curvature density has the exact form

$$K \det g = \partial_{\xi t} g_{\xi t} - \tfrac{1}{2}(\partial_\xi^2 g_{tt} + \partial_t^2 g_{\xi\xi}) + \mathcal{L}(g, \partial g),$$

with $\mathcal{L}$ smooth on uniformly positive matrices. The perturbation changes only g_{tt}. Therefore its only second-derivative contribution to this density is

$$\tfrac{1}{2}A_\tau\tau^2\chi\phi\cos(\tau\xi) + A_\tau\tau\chi'\phi\sin(\tau\xi) - \tfrac{1}{2}A_\tau\chi''\phi\cos(\tau\xi).$$

The first-derivative change is bounded by $CA_\tau(\tau + w_\tau^{-1})$; local Lipschitz continuity of $\mathcal{L}$, the Christoffel terms, and the rational nonlinearity bounds all such contributions. Replacing the multiplier of the leading curvature term by its value for g_* costs at most $CA_\tau^2\tau^2(\tau + w_\tau^{-1})$. This conservative estimate need not exploit additional cancellation specific to $H_{\xi\xi}$. The C^2 metric error costs $C\gamma_\tau$. The finite comparison residual costs Cw_τ^N, by (5).

The initial argument vectors agree for z, z^0. The C^8 height bound and the bounded finite comparison coefficients keep them $O(w_\tau)$ from that value. The mean-value segment is in a common compact jet region. At the initial solution jet, $Q_{g_*, z_{\xi\xi}} > 0$ by metric closeness and $d_0 < 0$. Continuity then gives a positive lower bound throughout the segment; the segment need not consist of solutions. Low metric and height derivatives bound its coefficient derivatives uniformly. Finally the numerator in (13) stays positive and its denominator stays nonzero with the sign opposite to H_{yy}. Connectedness gives a constant sign on the strip. This proves every claim. □

Theorem 4.2 (One-parameter pulse exclusion)**.** *Fix an integer $K \geq 8$, representing the largest low-order or neighborhood seminorm to be controlled. Suppose*

$$\tau w_\tau \longrightarrow 0, \qquad \tau^2 w_\tau \longrightarrow \infty. \tag{15}$$

Choose an integer $N > 2K$, use (12)*, and suppose $\gamma_\tau/(A_\tau\tau^2) \to 0$. Under the preceding cap, propagation, height-class and metric approximation hypotheses, there cannot be such a sequence of heights for arbitrarily large τ.*

For the power law $w_\tau = \tau^{-a}$, $1 < a < 2$, it suffices instead to take any integer $N > aK$. If $\gamma_\tau/(A_\tau\tau^2) = O(\tau^{-1/2})$, then $a = 3/2$ makes the normalized finite-certificate error $O(\tau^{-1/2})$, with class-dependent constants.

Proof. For each fixed $j \leq K$, differentiation of the pulse gives

$$\|\widetilde{g}_\tau - g_*\|_{C^j} \leq C_j A_\tau(\tau + w_\tau^{-1})^j.$$

Under (15), eventually $\tau < w_\tau^{-1} < \tau^2$; hence $N > 2K$ makes all these norms tend to zero. The power-law condition $N > aK$ proves the second version. Uniform low-order coefficient and hyperbolicity bounds now follow exactly as in Lemma 4.1, not by substituting a varying parameter into a fixed-parameter little-oh assertion.

The cap and propagation hypotheses bound the spatial derivatives of z_t at both edges. Lemma 2.1 does the same for z_t^0 at the terminal edge. Thus T_N in Theorem 3.1 is bounded uniformly. Also $K_\tau \leq C(1+\tau)$, $H_w \leq C$, and $J_\tau \geq \|\chi\|_2^2/4 > 0$ for large τ. Let $\Xi_\tau = A_\tau w_\tau \tau^2$ and put

$$\rho_\tau = \tau^{-1} + (\tau^2 w_\tau)^{-1} + A_\tau(\tau + w_\tau^{-1}) + \frac{\gamma_\tau + w_\tau^N}{A_\tau\tau^2}. \tag{16}$$

Since the spatial interval has bounded length, (14) yields $\epsilon/\Xi_\tau \leq C\rho_\tau$. Dividing (9) by Ξ_τ therefore gives the necessary inequality

$$0 < c_* \leq C\left[\frac{1}{\tau^2 w_\tau} + \rho_\tau + w_\tau(1+\tau)(1+\rho_\tau)\right]. \tag{17}$$

The first term includes the terminal moment: $\tau^{-N}T_N/\Xi_\tau = T_N/(\tau^2 w_\tau)$. Each term tends to zero. In particular

$$\frac{w_\tau^N}{A_\tau\tau^2} = \tau^{-2}(\tau w_\tau)^N \longrightarrow 0.$$

This contradicts (17).

For $w_\tau = \tau^{-a}$ the two largest width terms are τ^{1-a} and τ^{a-2}. Their slower decay has exponent $\min(a-1, 2-a)$, maximized at $a = 3/2$. All other displayed terms have at least this decay there when $K \geq 8$, $N > aK$, and the stated metric error holds. This optimizes this sufficient error budget only, not every possible oscillatory construction. □

Example 4.3 (A concrete schedule)**.** For a neighborhood requiring derivatives only through order eight, take $a = 3/2$, $N = 14$, and $\|g_\tau - \widetilde{g}_\tau\|_{C^\tau} \leq \tau^{-\tau}$. The pulse tends to zero in C^8 at least as τ^{-2}. Its main moment is of size $\Xi_\tau = \tau^{-27/2}$, while both competing relative width costs are $\tau^{-1/2}$. The comparison has degree fifteen, and its terminal moment uses fourteen spatial integrations by parts. These high derivatives come from the declared cap/propagation hypotheses; they do not follow from the C^8 bound alone. There is no explicit uniform numerical frequency threshold without bounds for those constants.

The general window (15) also permits, for example, $w_\tau = \tau^{-1}/\log\tau$ or $w_\tau = \tau^{-2}\log\tau$ for large τ. The first has a slowly decaying spatial-transfer cost, the second a slowly decaying one-derivative curvature cost. The source's fixed-width-ratio choice $w = \delta/\tau$ is not covered by (15), but remains valid: it makes the transfer cost a small fixed constant, after which the remaining errors vanish in its ordered limit. The new proof is an alternative regime, not a correction of that valid parameter order.

Example 4.4 (Why a small transfer cost is needed)**.** In the constant-coefficient equation $v_{tt} - v_{\xi\xi} = A\tau^2\phi(t/w)\cos(\tau\xi)$ on the circle of length 2π, choose an integer $\tau \geq 1$, $w = \pi/\tau$, and

$$\phi(s) = \rho(s - 1/2) + \rho(s + 1/2), \quad 0 \leq \rho \in C_c^\infty((-1/4, 1/4)), \quad \rho \not\equiv 0.$$

Its Fourier integral at frequency π is zero because it is $2\cos(\pi/2)\widehat{\rho}(\pi)$. The zero-data solution is $v = b(t)\cos(\tau\xi)$, with

$$b(t) = A\tau \int_{-w}^{t} \sin(\tau(t - s))\phi(s/w)\,\mathrm{d}s.$$

Both $b(w)$ and $b'(w)$ vanish. Thus nonnegative temporal forcing alone need not produce a nonzero terminal trace: spatial dynamics can cancel it when τw is not small. This is a wave-model negative control, not a metric satisfying all source hypotheses or a counterexample to its theorem.

5 Orientation and accumulation are separate steps

The height class has to be reached from an arbitrary hypothetical immersion. The source does not assume a uniform Hessian lower bound for all immersions. It first restricts a neighborhood of the origin until projection onto the tangent plane is a diffeomorphism, so the immersion is a graph.

In a disk with $K < 0$ inside and $K = 0$ on its boundary, the second fundamental form cannot vanish on the whole boundary. Indeed, the graph Hessian would then vanish there; along the connected boundary its gradient would be constant and its value affine. Subtracting that affine function produces a nonzero function zero on the boundary, with negative Hessian determinant inside. An interior extremum contradicts this. At a boundary point b where the second fundamental form is nonzero, it therefore has rank one. This is the source's elementary boundary-saddle argument, not a finite-jet obstruction.

Proposition 5.1 (A quantitative orientation margin)**.** *In the fixed Euclidean parameter coordinates let $H_b = \lambda uu^T$, $|u| = 1$, $\lambda \neq 0$. Take a rotated first axis making angle α with the kernel of H_b. If another symmetric matrix H obeys $\|H - H_b\|_{\mathrm{op}} \leq \eta|\lambda|$ and $\eta < \cos^2\alpha$, then*

$$|H_{yy}| \geq |\lambda|(\cos^2\alpha - \eta), \qquad \left|\frac{H_{xy}}{H_{yy}}\right| \leq \frac{|\sin\alpha\cos\alpha| + \eta}{\cos^2\alpha - \eta}. \tag{18}$$

Thus a ratio below q_* is certified whenever

$$\eta < \frac{q_*\cos^2\alpha - |\sin\alpha\cos\alpha|}{1 + q_*}.$$

Proof. The rank-one entries are $\lambda\cos^2\alpha$ and $\lambda\sin\alpha\cos\alpha$, up to the choice of sign of an axis. The operator-norm perturbation bounds the absolute error of each entry by $\eta|\lambda|$. The triangle inequality gives both estimates; multiplying by the positive denominator gives the final criterion. □

For the source net $|\alpha| \leq 1/1000$, taking $\eta = 1/200$ gives a ratio less than $1/100$, by $|\sin\alpha\cos\alpha| \leq |\alpha|$ and $\cos^2\alpha \geq 1 - \alpha^2$. Independently, continuity of the fixed-normal height at b makes $|dz|_g^2 < 1/2$ on a

sufficiently small neighborhood. The patch must lie wholly in that neighborhood. Its size depends on this immersion, λ, and continuity moduli.

For an affine patch map $\Phi(x) = c + \ell Rx$, the actual metric pullback is $\ell^2 R^T (g \circ \Phi) R$; curvature is the scalar $K_g \circ \Phi$. The Hessian pulls back with the same ℓ^2, preserving the ratio in (18). Pushing a model tensor back costs ℓ^{-m-2} in a derivative of order m. No rescaling of curvature or replacement of this tensor cost by a coordinate norm is permitted.

The source places full obstructed patches of every net orientation approaching every boundary point of each negative disk. Those disks in turn approach the origin. It chooses each perturbation small in finitely many seminorms, with summable tails at every fixed derivative order. This gives a smooth metric retaining the required disk and patch properties. Applying the scalar obstruction to the selected whole patch is then a contradiction for that immersion. The countable bounded classes and Baire theorem remove the scalar bounds; a finite list of probes cannot replace them.

5.1 Flat jets do not determine smooth germs

The background metric is $\mathrm{d}X^2 + f^2\,\mathrm{d}Y^2$, where $f_{XX} = -K_0 f$, $f(0,Y) = 1$, $f_X(0,Y) = 0$. If K_0 has all derivatives zero at the origin, the complete Leibniz expansion of this ODE shows the same for $f - 1$. The summably differentiated tensor series preserves that jet. This part of the source argument is valid only because smoothness and derivative convergence have already been proved.

Conversely, the smooth function $\exp[-1/(X^2 + Y^2)]$, extended by zero at the origin, is nonzero arbitrarily nearby and has every derivative zero there. A Taylor jet, even an infinite one, therefore cannot determine a general smooth metric germ. Neither this classical example nor a finite exact jet computation proves the source's nonimmersion claim.

6 Comparison, costs and evidence boundaries

The local increment is the finite inequality (9) and its controlled width extension, including all differentiated coefficients, initial/terminal defects and Taylor errors. The source already provides the curvature forcing identity, short-strip energy method and fixed-δ contradiction. Those are credited inputs, not discoveries here. The exact flux speed and rank-one margin are supporting classical computations.

The advantage is that each possible failure of a finite obstruction is visible: an inadequate signed curvature moment, an unresolved terminal norm, a large PDE residual, loss of hyperbolicity, or a strip wide enough for transfer to absorb the forcing. The power schedule eliminates one ordered small-constant choice, but increases transverse derivative costs and forces a correspondingly large amplitude exponent. It supplies neither a measured computational speedup nor a universal frequency valid across patches. The order-$\tau^{-1/2}$ relative budget is not an optimal nonimmersion exponent or a new global theorem.

The original TR208 proof [6] supplies exact multi-index Leibniz rows, conditional bandwidth rather than unconditional triangularity, and a Taylor remainder with a derivative supremum. The original jet chapter [7] explicitly carries differentiated remainders and the omitted smooth closure term. Their finite polynomial recovery assertions apply on a declared polynomial class. They do not turn the Darboux equation into an invertible finite system, and their integral-kernel stability results are not hyperbolic propagation theorems.

One possible extension is to enclose the norms and signs in (9) by validated local computations, together with an independent analytic cap bound. Another is to exploit structural cancellations in the one-derivative curvature error before optimizing w; the present interval $1 < a < 2$ is sufficient, not proved necessary. Robust cap estimates for approximate Darboux heights would require differentiated

residuals of every order used in the induction. They are not supplied merely by a small uniform residual or by sampled derivative rows.

6.1 Public literature status

The October 9, 2026 primary-source refresh distinguishes regularity and publication status. Nadirashvili and Yuan's 2002 arXiv abstract [2] claims a smooth nonpositive curvature counterexample. Their 2008 published paper [3] states a $C^{2,1}$, nonnegative-curvature counterexample with no C^2 local embedding, and its introduction still describes the arbitrary smooth problem as open. The assigned source itself leaves this historical discrepancy unresolved. We do not infer a retraction, correction, or settled smooth theorem from these records.

Khuri's 2009 primary paper [4] reduces Darboux nonsolvability to a related equation and distinguishes general Monge–Ampère counterexamples from metric-compatible ones. Han and Khuri [5] give positive results under finite-order and transverse-zero-set assumptions. The original mixed-sign source construction does not remove those distinctions. Access here covered primary abstract/version records and selected introductory PDF text, not complete historical proofs. Requested web PDF screenshots were unavailable, so no visual historical-PDF proof review is claimed.

6.2 Reading and checks

All eleven assigned public text files were freshly read in bounded literal displays: 1,867 lines, including all four mathematical modules. The original TR208 was read at lines 143–380; the original jet chapter at 513–583, 747–852, 897–934 and 1211–1305. Other portions and included references were not read. PRE hashes were persisted before these displays. The owned ledger records POST equality and exact unread complements; parent and prior report reads are not counted.

The owned executable checks pass 669 exact cases with 1,409 assertions, using rational polynomial arithmetic and exact finite identities; there are no floating-point cases. They exercise exact coefficient differentiation, Taylor comparison, energy/flux algebra, pulse scaling, orientation margins and explicit negative controls. They test manufactured finite data, not cap regularity, Baire category, smooth accumulation, or an arbitrary immersion. Exact cases and any numerical diagnostics are reported separately in the final receipt.

References

[1] OpenAI, *A Smooth Metric with No Local Isometric Immersion into Three-Space*, September 24, 2026. Pinned source tree. Assigned text read; global validity not independently certified.

[2] N. Nadirashvili and Y. Yuan, *Counterexamples for Local Isometric Embedding*, arXiv:math/0208127v1, August 16, 2002. Abstract and version metadata inspected.

[3] N. Nadirashvili and Y. Yuan, *Improving Pogorelov's isometric embedding counterexample*, Calc. Var. PDE 32 (2008), 319–323. Author-hosted journal paper. Introductory text, including Theorem 1.1 and the open-problem discussion, inspected; full proof not audited.

[4] M. A. Khuri, *On the local solvability of Darboux's equation*, DCDS Supplement (2009), 451–456. Publisher record; author-hosted text. Abstract and initial introductory text inspected.

[5] Q. Han and M. Khuri, *On the local isometric embedding in $\mathbb{R}^3$ of surfaces with Gaussian curvature of mixed sign*, Comm. Anal. Geom. 18 (2010), 649–704. arXiv:1009.6214v1. Abstract and version record inspected.

[6] M. Subbarao, original confidential TR208, *Multidimensional variable-coefficient operators*, local file in `TechReps/FinalTechReps/TRs`, lines 143–380. Full locator and hashes in the local read ledger.

[7] M. Subbarao, original confidential *Local derivative-jet reduction*, `RamNonlinearIE-PDE/TR1-lin-nonlin-rev2`, `sec-03-jets.tex`. Scoped theorem and proof reads listed in Section 6.2; unlisted includes unread.